AI Powered keyword extraction from problem solving image

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Motivations

For students who prepare Korean SAT, it is important to **record and revisit unsolved questions** but it could be tedious or daunting.

Additionally, in our existing mobile application, many students were not using the feature for recording unsolved questions.

A new feature that **extracts keywords from question image** and **organize them by keyword** could help students more easily record and revisit unsolved questions.

AI, LLM Application

OCR

Pix2Text OCR model is implemented to extract Korean texts and mathematical equations at the same time.

Sentence Transformer

The extracted text is processed by **Sentence Transformer** which selects **top-k keywords** related to the given question, from a predefined set of 20 keywords.

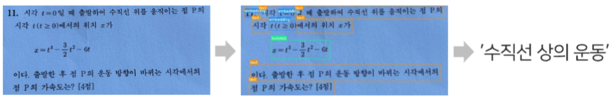


Figure 1. Keyword extraction process

Prototype architecture

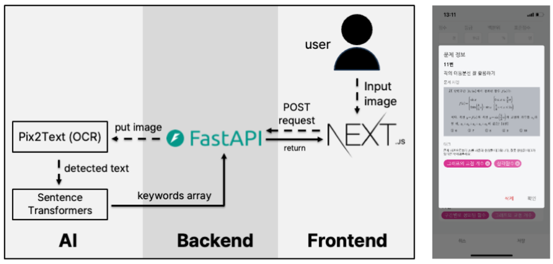


Figure 2. Prototype architecture

- We developed the frontend using **Next.js**, which communicates with the server using API.
- The server is based on **FastAPI** and runs on a cloud instance.

User study

Survey questions

	Before	After
Q1	나는 실전 모의고사 응시 후 틀린 문제/시간 내에 풀지 못한 문제에 대해 실감 앱으로 오답노트를 작성한다.	나는 앞으로 실전 모의고사 응시 후 틀린 문제/시간 내에 풀지 못한 문제에 대해 실감 앱으로 오답노트를 작성할 것이다.
Q2	나는 실전 모의고사 응시 후 틀린 문제/시간 내에 풀지 못한 문제를 풀기 위해 필요했던 핵심 아이디어가 무엇인지 고민해본다.	나는 앞으로 실전 모의고사 응시 후 틀린 문제/시간 내에 풀지 못한 문제를 풀기 위해 필요했던 핵심 아이디어가 무엇인지 고민해볼 것이다.
Q3	나는 실전 모의고사 응시 후 앱에 기록했던 오답 문항들을 이후 다시 복습한다.	나는 앞으로 실전 모의고사 응시 후 앱에 기록했던 오답 문항들을 향후 다시 복습할 것이다.
Q4	나는 수학 과목의 실전 모의고사에서 틀렸던 문제들을 주제별로 모아서 복습한다.	나는 앞으로 수학 과목의 실전 모의고사에서 틀렸던 문제들을 주제별로 모아서 복습할 것이다.

Table 1. Survey questions

Analysis

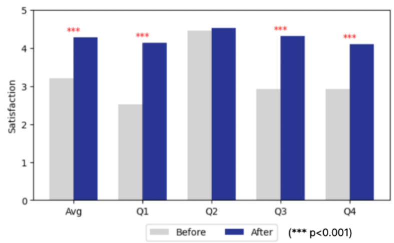
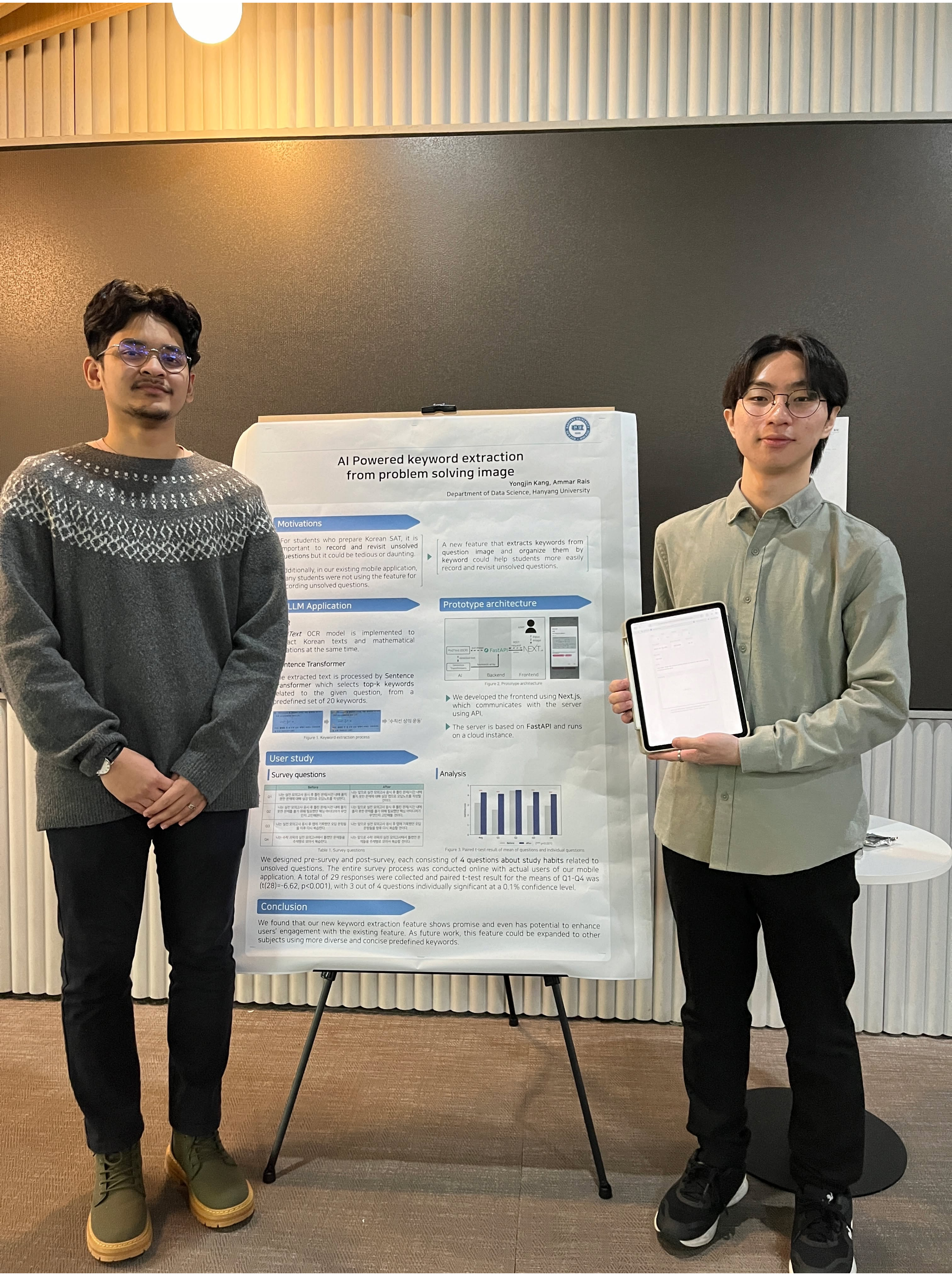


Figure 3. Paired t-test result of mean of questions and individual questions

We designed pre-survey and post-survey, each consisting of **4 questions about study habits** related to unsolved questions. The entire survey process was conducted online with actual users of our mobile application. A total of 29 responses were collected and paired t-test result for the means of Q1-Q4 was ($t(28)=-6.62$, $p<0.001$), with 3 out of 4 questions individually significant at a 0.1% confidence level.

Conclusion

We found that our new keyword extraction feature shows promise and even has potential to enhance users' engagement with the existing feature. As future work, this feature could be expanded to other subjects using more diverse and concise predefined keywords.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LLM Application
Prototype architecture
AI
Backend
Frontend
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We developed the frontend using Next.js, which communicates with the server using API.
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'수직선 상의 운동'
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User study
Survey questions
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